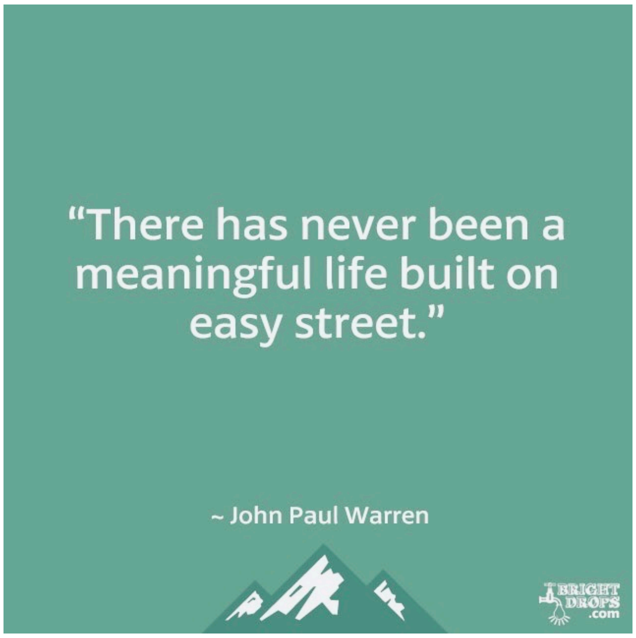
2D MATRICES

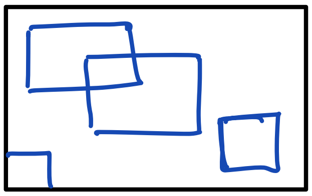


Good Evening

Today's Content

01. Introduction to submatrix
02. Submatrix sum queries
03. Sum of all submatrix sum
04. Maximum sum submatrix {Hint}

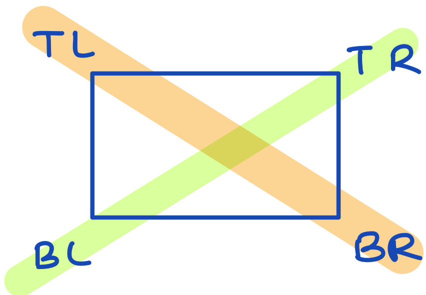
Submatrix : Part of a matrix is a submatrix



Note :- (a) Single ele is a submatrix

(b) Complete matrix is also a submatrix

Identify : Any submatrix contains 4 co-ordinates



No need for all 4 corners

→ TL & BR or

→ TR & BL

Example :- int mat[5][6];

	0	1	2	3	4	5
0	2	-1	3	2	1	3
1	3	2	6	2	6	7
2	10	9	8	2	2	1
3	4	-1	2	3	4	2
4	3	2	6	9	8	9

TR	BL	TL	BR
0, 2	2, 0	0 0	2, 2
1, 3	4, 1	1, 1	4, 3

Conclusion → Given {TL & BR} or {TR & BL}

→ submatrix can be identified

Generally → {TL , BR} for submatrix identification

Q1. Find sum of all elements in a given submatrix

Constraints

$1 \le N * M \le 10^6$

$1 \le mat[i][j] \le 10^6$

Example

	0	1	2	3	4	5
0	2	-1	3	2	1	3
1	3	2	6	2	6	7
2	10	9	8	2	2	1
3	4	-1	2	3	4	2
4	3	2	6	9	8	9

TL	BR		sum
(1,2)	(3,5)	→	45

TL	BR		
(0,3)	(2,4)	→	15

Idea → Iterate on the given submatrix & add all of the elements

```
long ssum (int [] [] mat, int r1, int c1, r2, c2 )
        TL                 BR
    long sum=0
    for ( i=r1 ; i <= r2 ; i++ )
        for ( j=c1 ; j <= c2 ; j++) {
            sum += mat [i] [j] ;
        }
    }
    return sum;
}
```

$TC = O(m*n)$
$SC = O(1)$

Q2. Given a matrix of size N * M . & Q queries

For each query, find sum of given submatrix

Note :- TL is top left & BR is bottom right

Constraints

$1 \le N*M \le 10^6$

$1 \le Q \le 10^5$

$1 \le mat[i][j] \le 10^6$

	0	1	2	3
0	2	-1	3	2
1	3	2	6	2
2	10	9	8	2
3	4	-1	2	3
4	3	2	6	9

Queries = 2

(T, L) (B, R)

01 (2,1) & (4,2) → 26

02 (1,1) & (3,3) → 33

Brute force → For every query, iterate on the submatrix & find the sum

$TC = O(Q*n*m)$

$SC = O(1)$

$TC = 10^5 * 10^6 * 10^6 = 10^{17}$

TLE

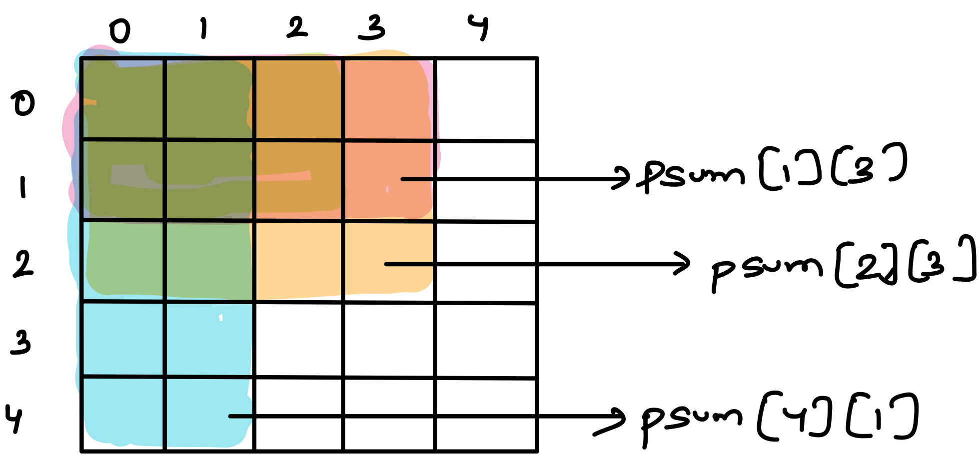
Idea 2 → psum approach

1D Array psum[i] = sum of all ele from 0 to i

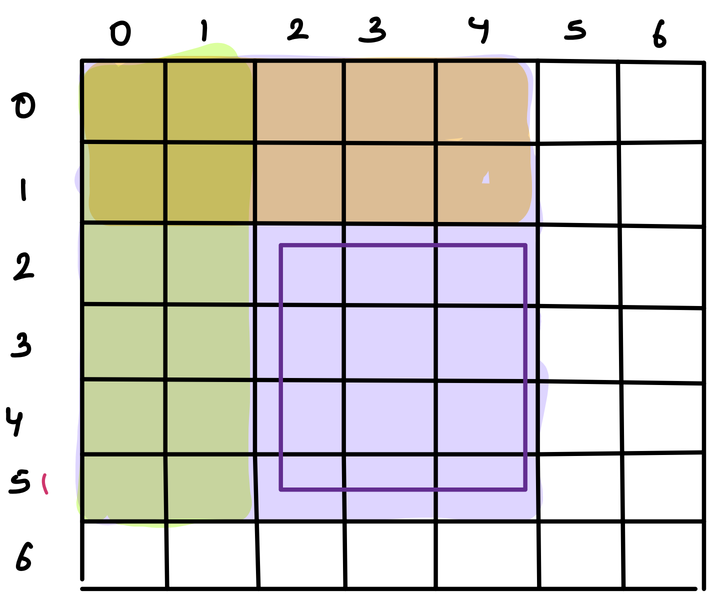
↓ Extending this for 2D matrix

Psum[i][j] = sum of all ele from (0,0)→(i,j)

Psum[4][4] =



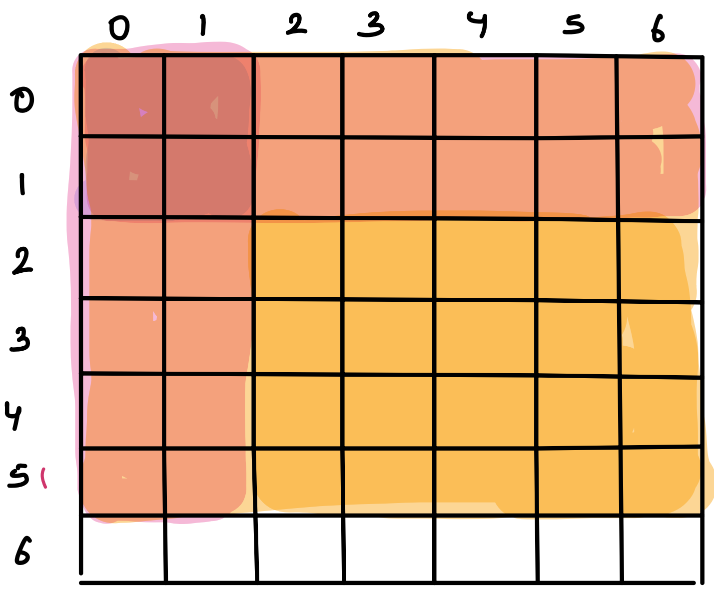
Assume psum[][]



Query

(2,2) → (5,4)

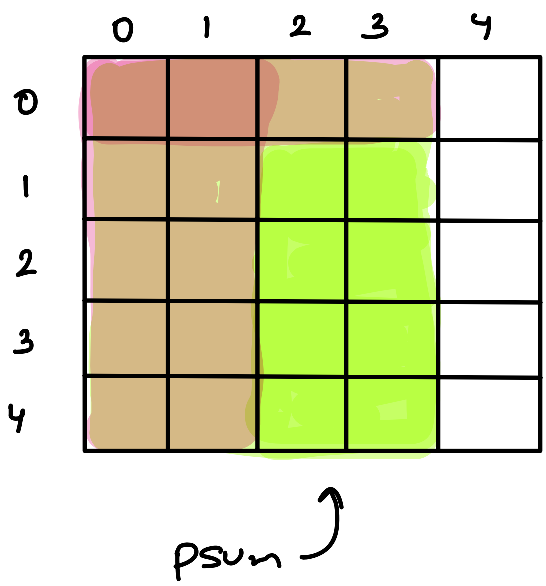
psum(5,4) − psum(1,4) − psum(5,1) + psum(1,1)



Query

(2,2) (5,6)

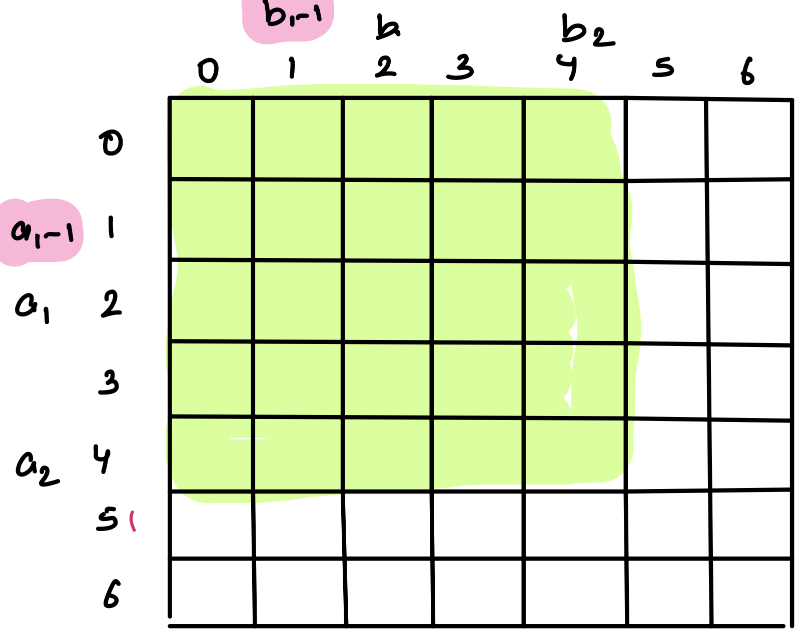
psum(5,6) − psum(1,6) − psum(5,1) + psum(1,1)



Query

(1,2) $\longrightarrow$ (4,3)

psum(4,3) − psum(0,3) − psum(4,1) + psum(0,1)



TL $\quad$ BR

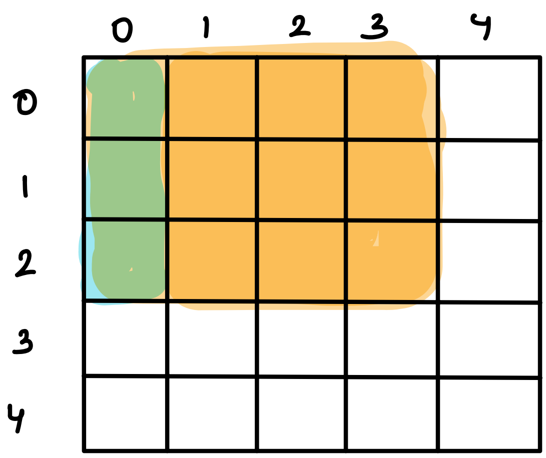
$(a_1, b_1) \quad (a_2, b_2)$

$$ans = psum(a_2, b_2) - psum(a_1-1, b_2) - psum(a_2, b_1-1) + psum(a_1-1, b_1-1)$$

```
ans = psum[a2][b2]

if (a1 > 0) {
    ans = ans - psum[a1-1][b2]
}

if (b1 > 0) {
    ans = ans - psum[a2][b1-1]
}
if (a1>0 && b1>0)
    ans = ans + psum[a1-1][b1-1]
}
```



psum

TL	BR
(0,1)	(2,3)

$$ans = psum[2][3] - psum[2][0]$$

* Calculate psum matrix

01. Copy all elements from given matrix to psum
02. Apply psum on every row
03. Apply psum on every col

	0	1	2
0	a_0	b_0	c_0
1	a_1	b_1	c_1
2	a_2	b_2	c_2

mat[ ][ ]

	0	1	2
0	a_0	b_0	c_0
1	a_1	b_1	c_1
2	a_2	b_2	c_2

psum [ ] [ ]

Apply psum on every row

a_0	a_0+b_0	$a_0+b_0+c_0$
a_1	a_1+b_1	$a_1+b_1+c_1$
a_2	a_2+b_2	$a_2+b_2+c_2$

Apply psum on every col

a_0	a_0+b_0	$a_0+b_0+c_0$
$a_0 + a_1$	a_0+b_0 + a_1+b_1	$a_0+b_0+c_0$ + $a_1+b_1+c_1$
a_0 + a_1 + a_2	a_0+b_0 + a_1+b_1 + a_2+b_2	$a_0+b_0+c_0$ + $a_1+b_1+c_1$ + $a_2+b_2+c_2$

psum[ ] [ ]

Mat

	0	1	2	3
0	3	2	4	1
1	-1	4	3	2
2	2	7	6	3

final psum mat

psum

	0	1	2	3
0	3	5	9	10
1	2	8	15	18
2	4	17	30	36

	0	1	2	3
0	3	2	4	1
1	-1	4	3	2
2	2	7	6	3

psum

Apply psum on rows

	0	1	2	3
0	3	5	9	10
1	-1	3	6	8
2	2	9	15	18

Apply psum on col

	0	1	2	3
0	3	5	9	10
1	2	8	15	18
2	4	17	30	36

* Construct prefix sum matrix

```
long psum[ ][ ] = new long[n][m]
```

01. Step 1 → Copy the ele

```
for (i=0; i<n; i++)
|   for (j=0; j<m; j++) {
|   |   psum[i][j] = mat[i][j]
|   }
}
```

02 step 2 → Cumulative sum for rows

```
for (i=0 ; i<n; i++)
    long sum = 0
    for ( j=0; j<m ; j++) {
        sum = sum+ psum[i] [j]
        psum[i][j]= sum
    }
}
```

03. Step 3 → Cumulative sum for cols

```
for ( j=0; j<m ; j++) {
    long sum = 0
    for (i=0 ; i<n; i++)
        sum = sum+ psum[i] [j]
        psum[i][j]= sum
    }
}
```

⇒ for loop for queries now

$TC = O(n*m + q)$

$SC = O(n*m)$

Q2. Given a matrix of size $N*M$. Calculate **sum of all submatrix sums**

$$\begin{bmatrix} 3 & 1 \\ -1 & -2 \\ 2 & 4 \end{bmatrix} = \left\{ \begin{array}{l} [3] \quad [3, 1] \quad \begin{bmatrix} 3 & 1 \\ -1 & -2 \end{bmatrix} \quad \begin{bmatrix} 3 & 1 \\ -1 & -2 \\ 2 & 4 \end{bmatrix} \quad \begin{bmatrix} 3 \\ -1 \end{bmatrix} \quad \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix} \\ [1] \quad \begin{bmatrix} 1 \\ -2 \end{bmatrix} \quad \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} \quad [-1] \quad [-1, -2] \quad \begin{bmatrix} -1 \\ 2 \end{bmatrix} \quad \begin{bmatrix} -1 & -2 \\ 2 & 4 \end{bmatrix} \\ [-2] \quad \begin{bmatrix} -2 \\ 4 \end{bmatrix} \quad [2] \quad [2, 4] \quad [4] \end{array} \right\}$$

Sum = 36

<u>Idea</u> → Consider all the submatrices, get the sum & add them to total sum

$$\text{Total no. of submatrices} = \frac{N(N+1) * M(M+1)}{4}$$

<u>Idea 2</u> → Contribution technique

$\Rightarrow 3*6 + 1*6 + (-1*8) + (-2*8) +$
$(2*6) + (4*6)$

$\Rightarrow 18 + 6 - 8 - 16 + 12 + 24 = \boxed{36}$

* How many times is a particular ele coming in all submatrices

	0	1	2	3	4
0	TL	TL	TL		
1	TL	TL	TL BR	BR	BR
2			BR	BR	BR
3			BR	BR	BR

How many times will (1,2) be present in submatrices

$$\underbrace{TL}_{\text{Possible options for TL}} \ * \ \underbrace{BR}_{\text{Possible options for BR}}$$

$6 * 9 \Rightarrow 54$ submatrices

	0	1	2	3	4
0	TL	TL	TL	TL	
1	TL	TL	TL	TL	
2	TL	TL	TL	TL BR	BR
3				BR	BR
4				BR	BR

n*m

No. of submatrices where (2, 3) is present?

TL * BR

12 * 6 = 72

No. of times a particular ele is going to contribute → $(i+1)(j+1) * (n-i)(m-j)$

Pseudocode →

$TC = O(n*m)$
$SC = O(1)$

```
ans = 0
for (i = 0; i < n; i++)
|   for (j = 0; j < m; j++)
|   |   long count = (i+1)(j+1) * (n-i)(m-j);
|   |   ans = ans + ar[i][j] * count
|   3
3
return ans;
```

Q3. Given row wise & col wise sorted matrix, find

maximum submatrix sum

	0	1	2	3
0	-20	-16	-4	8
1	-10	-8	2	14
2	-1	6	21	30
3	5	7	28	42

To find Maximum submatrix sum

↓

Maximum ele

BR = mat [n-1] [m-1]

↓

Explore all the possibbilities for the TL corner

Pfsum

	0	1	2
0	-15	-14	-13
1	-12	-11	-7
2	-10	-9	-3

BR = (2,2)

TL = (0,0) (1,0) (2,0)
(0,1) (1,1) (2,1)
(0,2) (1,2) (2,2)

Doubts

$A = \{-2, 1, -3, 4, -1, 2, 1, -5, 4\}$

sum = 0   -2   . . .

ans = $-\infty$ -2   . . . {TODO}